

VISUAL INFORMATION SYSTEM SOFTWARE DESIGN CRITERIA

BY

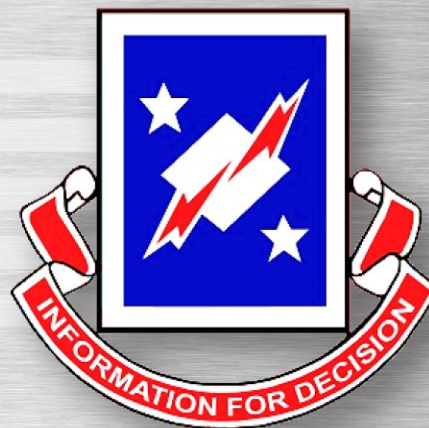
COMMAND AND CONTROL GROUP

MARCH 2018

FORT HUACHUCA ENGINEERING DIRECTORATE

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EXECUTIVE SUMMARY

The Department of Defense (DoD) and U.S. Army have defined the goal of designing and deploying systems that are compatible in the production, use, and electronic exchange of information. These systems will provide seamless integration of Visual Information System (VIS) components, interoperate within the VIS environment and with external systems, and maximize the potential of new and improved technologies to provide support to the end users. In support of their common goal, the DoD and U.S. Army have developed guidelines and standards for system development and acquisition. The U.S. Army Information Systems Engineering Command (USAISEC), in turn, developed this VIS Software Design Criteria as a way to assist system developers, engineers, and managers in designing a standardized VIS for Command and Control Facilities (C2Fs) that support the DoD and Army goals.





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VISUAL INFORMATION SYSTEM SOFTWARE DESIGN CRITERIA

1.0 INTRODUCTION

1.1 Scope

The goal of this document is to supplement customer Visual Information System (VIS) software design requirements defined in project System Design Plans (SDPs) or other requirements documents, and to provide the software designer with U.S. Army Information Systems Engineering Command's (USAISEC's) expectations for VIS Software design, to include, defining VIS software deliverables for the Engineering Installation Plan (EIP) submissions. This document promotes long term software maintainability and enables simplified user training and documentation. Methodologies presented in this document are guidelines with exceptions being approved at the Project Leader level, and based upon unique project requirements.

1.2 Document Organization

This document is divided into the following sections: Introduction, VIS System Overview, Control Touch Panel Graphical User Interface (GUI) Test Plans, Operator Training Guide, Maintenance Test Routines, Operator/User Macro Capability, and Software Development Milestones. This document also includes the following:

- Appendix A. References
- Appendix B. Sample Test Plans
- Appendix C. VIS Requirements Matrix
- Glossary. Acronyms and Abbreviations

2.0 VIS SYSTEM OVERVIEW

2.1 Background

The VIS encompasses the Briefing and Display (B&D) system, the Video Teleconferencing (VTC) system, and the Audiovisual (A/V) Distribution Subsystems (VDSs), to include the A/V control system. The VIS software solution, part of the control system, provides the capability to control the distribution of A/V signals of various classifications, and provide seamless and secure transfer of information between Command and Control Facilities (C2Fs). The VIS software development shall have the flexibility to operate in multi-level environments. Switching the VTC endpoint between different classifications of networks should be an intricate, closely evaluated procedure. The VIS software must follow a very strict set of sequenced events to ensure information does not leak from one network to the other. Most of these steps are automated through the A/V control system and should require only a minimal amount of actual user interaction. The VIS should rely on as little user intervention as possible to reduce the possibility of human error. The SDP captures the engineering efforts needed to design the VIS and associated subsystems. An EIP, separate from this document, will detail the design and implementation of each system described in the SDP and will be approved by each stakeholder prior to commencing any implementation.





2.2 VIS Requirements Matrix

The VIS requirements matrix shown in Figure 1 (Appendix C contains the full Excel version), provides a consistent and streamlined development process, and serves as a software and User Interface (UI) development map. This map enables the software developer to accurately estimate the needed software features such as, UI pages, pop-up pages, device drivers, and work flows. Using this approach when designing the software and UI architecture, the developer can quickly grasp the overview of the projects needs and construct an inventory of needed elements.

	Command Suite	Senior Planning Room	Executive Conference	Operations Center	OC Planning Rooms	NOC Office	Small Conference
Video Display							
Knowledge Wall							
LCD/LED Screen							
Integrated TP Control - 10"							
Video Service							
Satellite/CATV							
VCR/DVR							
DVD							
BluRay							
internal inputs - IP							
external inputs - IP							
external outputs							
VTC Capability							
NIPR							
SIPR							
Other							
Workstations							
NIPR							
SIPR							
JWICS							
Other							

Figure 1. Sample VIS Requirements Matrix

2.3 Functional Description/System Capabilities

2.3.1 A/V Control System

The functional requirement for an A/V control system is to maintain control of the inter-room A/V operations, as well as central management of A/V assets. The A/V control system shall have the capability to control A/V assets within the conference room and remotely from the A/V control room. The A/V control system equipment shall be standardized across conference rooms to simplify maintenance, where feasible. A/V system users in designated conference rooms shall select A/V sources and destinations using a touch panel or a laptop computer. Each conference room will have its own touch panel to allow local control of the room. The control room shall have the capability to remotely control all A/V equipment in each conference





area. The control system programming code shall be line-commented and easily understood to facilitate future modifications.

The A/V control system must implement the programming code in a single language, to prevent security concerns with the use of several different language combinations. The source code and compiled versions, with all extensions and attachments, must be provided for U.S. Government (USG) review.

2.3.2 Multi-level Security

This section describes the mechanism for the operator/user to set the security level of the room. This mechanism will cause a restriction within the main A/V switching system, so that only devices allowed for use, will be available for selection on the UI and able to route through the A/V matrix switch.

Any source device rated at a security level higher than the current room security level, shall be restricted from use. Any device rated below the current room security level that is capable of transmitting, shall be restricted from use.

The device routing buttons on the UI will be phantom or dimmed to indicate when not legal or available to be used, at the current room security level. Additionally, the phantom or dimmed buttons will not respond to button presses when not legal or available for use, at the current room security level. The UI shall clearly indicate the current room security level.

2.3.3 User Authentication

User authentication shall be used to restrict multi-level security access and is used to restrict non-authorized users from operating a room. The user authentication system shall be tied to the core multi-level security system, which restricts the use of certain hardware devices and possibly tie-lines depending upon the selected security level.

Certain authenticated users (Super Users), are allowed to select any room security level while others may be restricted to one of the mid security levels. Others will be restricted to operate only at Unclassified levels.

This authentication process shall make use of an editable file. This file will be stored on the local control processor or on an external device like a headless processor. Software will be provided to allow for edits, backups, and updates to the authentication file.

User authentication shall present the operator/user with a log-in dialog where they will enter a pin code, password, or both. The authentication process will verify user credentials against the authentication file and grant or deny access to select a room security level. Some rooms will require different users to have access to different devices; in this case, the authentication process will either initiate UI page flips or cover/uncover particular hardware assets. There will also be methods tied to room shutdown procedures that remove the current authentication and require re-authentication to restart the room.

2.3.4 UI Components

Software components, used to construct UI pages and build the core software, will be closely linked to the VIS requirements matrix as described above. UI elements will be consistent throughout all touch panels and UIs.





- Color palette and themes
 - o Background colors, themes, text colors, and banner colors shall be consistent across all rooms.
- Logos
 - o Unit logos shall be obtained from the individual Units. (Integrator logos are NOT allowed to be placed anywhere within the UI.)
- Lists
 - o List background colors shall ensure the list items are legible in all lighting levels.
 - o Scroll methods and scroll button locations shall be the same for every list.
- Buttons and button styles
 - o Button styles shall have visual feedback to indicate state or status wherever possible.
 - o Page select buttons shall be of a different shape than function control buttons, to give the user visual clues as to the meaning of the button.
- Label and hint types
 - o Labels and hints shall use colors that will be legible against backgrounds in all lighting levels.
- Pop-up page styles and borders
 - o The developer shall limit the use of pop-up or sub-pages as much as possible. The goal is to present a smooth operator work flow and eliminate a confusing array of page configurations. The user shall be able to easily understand where they are within a user work flow.
 - o When used, pop-up or sub-pages shall have a clear method of dismissal to allow the operator to change their mind and exit back to the previous state.
 - o When used, pop-up or sub-pages shall NOT have a transparent background that allows for any background page to show thru the surface of the pop-up or sub-page.
 - o When used, all pop-up or sub-pages shall have a border to delineate the boundary of the pop-up or sub-page.
- Confirmation dialogs and styles
 - o Confirmation dialogs shall be consistent throughout the UI and shall be distinctly different from all other dialogs.
- Developers shall provide Common Device Drivers for the following categories:
 - o VTCs
 - o Routing switches
 - o Infrared (IR) Device Files format





- o Display control methods

2.3.5 Hardware/Software Configuration Control

- IR File Version
 - o Each IR file used in the development of the software shall be listed by file name, the room number, and the software version the IR file was used in.
 - o The actual file shall be archived along with the room source code as part of the final software deliverable.
- Hardware Firmware Version (A/V Hardware)
 - o Each hardware device shall have its current firmware version listed in the version control document for area or location.
 - o A copy of the firmware file shall be archived along with the room source code as part of the final software deliverable.
- Library Version
 - o The developer shall provide the Software Developer Library versions used in the development (if AMX is the development platform).
 - o A copy of all library files shall be archived along with the room source code as part of the final software deliverable.
- Database Version
 - o The developer shall provide a Software Developer tools database and device database used in the development (if Crestron is the development platform).
 - o A copy of the device database and user database files will be archived along with the room source code as part of the final software deliverable.
- UI File Version
 - o Each UI file version shall be dated with the Tested and Accepted (T&A) version. Example: RoomXXX-UI-10-23-17.yyy
- Processor Core File Version
 - o Each Core file version shall be dated with the T&A version. Example: RoomXXX-Core-10-23-17.yyy

2.3.6 GUI

The GUI shall provide a presenter or operator with a clear, well organized, and user-friendly Visual Interface (VI) to control all A/V Inputs/Outputs (I/Os) in a given room. It must accurately show the current state of all devices and audio levels in the room. The interface pages must be organized and presented in a neat and logical manner, and shall be coordinated with the customer. The GUI must enable a user to temporarily downgrade the conference room or work area to a lower classification or sanitize the room (blank all displays) when uncleared personnel are present. The A/V signs located inside the conference room and above the entry door(s) shall display the classification status indicator.





The sample UI layout in Figure 2 depicts a number of key concepts and a key work flow. The main goal is to minimize the amount of operator/user interaction, eliminate multiple page flips and pop-up pages, while accomplishing the most work.

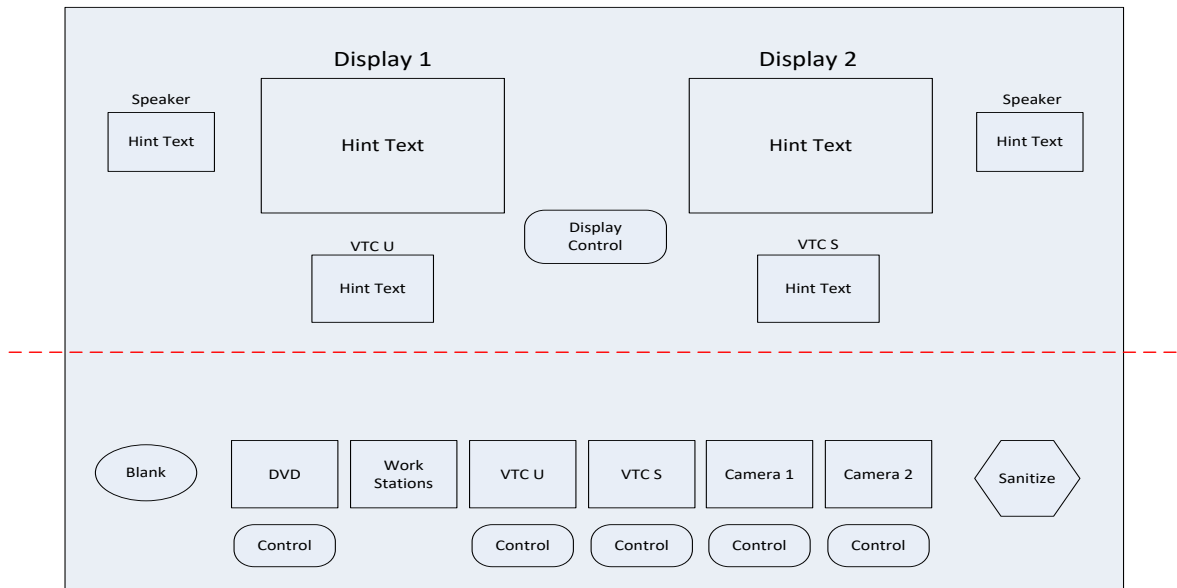


Figure 2. Sample GUI Layout - Main Page

The red dotted line across the center of the image is an important division in the work flow. Every feature below the red dotted line is considered to be an input or source to the room, and every feature above the red dotted line is considered to be an output to or from the room. An example work flow follows: to route the Digital Versatile Disc (DVD) to Display 2, the operator/user will first press the DVD button and then press the Display 2 button to route the video, the hint text within the Display 2 button will update to indicate “DVD” as confirmation to the operator/user that the route has taken place (this also supports the “at a glance” situational awareness that operator/users rely on), they will then press one of the two speaker button to route the audio, the speaker hint text is updated as well. This type of modeless UI is the most flexible; in that it allows the operator/user to quickly route any signal without having to leave the control page. If the operator/user wishes to control the DVD, they will simply press the control button beneath the DVD button and a DVD control page will pop up. To route a source to multiple outputs, the operator/user does not need to reselect the desired input device again, the source is current until another has been selected. Additionally, to reduce the level of confusion on the part of the operator/user, button styles shall remain consistent throughout the UI. Buttons that perform page flips will be a different shape or style than buttons that perform functions. The display control and the individual control button are the same style. Blank and Sanitize are different from each other and page flip or pop-up page buttons. The workstation button automatically pops up a partial page where the operator/user will continue with the selection. Since this function serves as a source selection, the button shall appear the same as other source selection buttons.





The sample UI layout in Figure 3 is a sample of a multiple-input selection pop-up page. When a source has more items that can fit on the main page, it is acceptable to have an automatic pop-up page to allow for selections. Note the use of the “Auto Close” checkbox. This is a toggle function the operator/user will change to allow for an automatic dismissal of this type of selection dialog when a single item has been chosen. Un-checking this item will keep the selection dialog on screen until the operator/user manually dismisses it with the close button. The close button is always active in the event the operator/user changes their mind or needs to back out from an accidental selection on the main page.

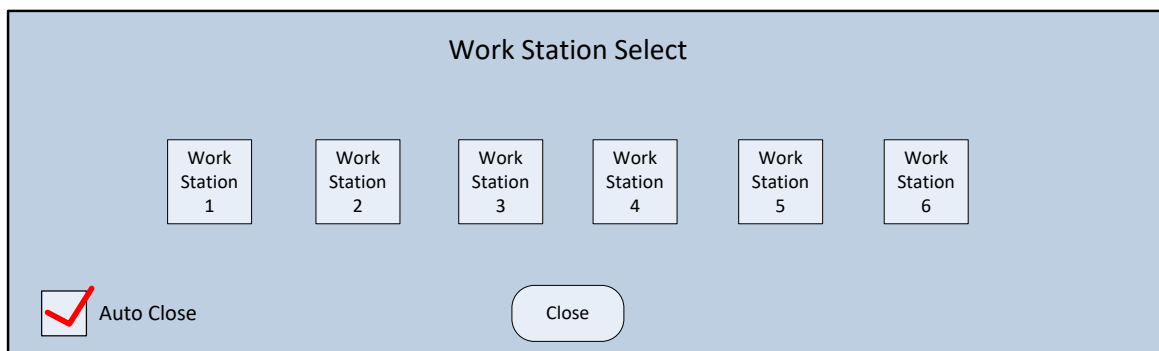


Figure 3. Sample GUI Layout - Pop-up Page

The sample user interface layout in Figure 4 is a sample of a device operations page. This device operations page pops up when the operator/user presses the control button beneath the DVD button on the main page. This device operations page stays on the screen until the user dismisses it with the close button.

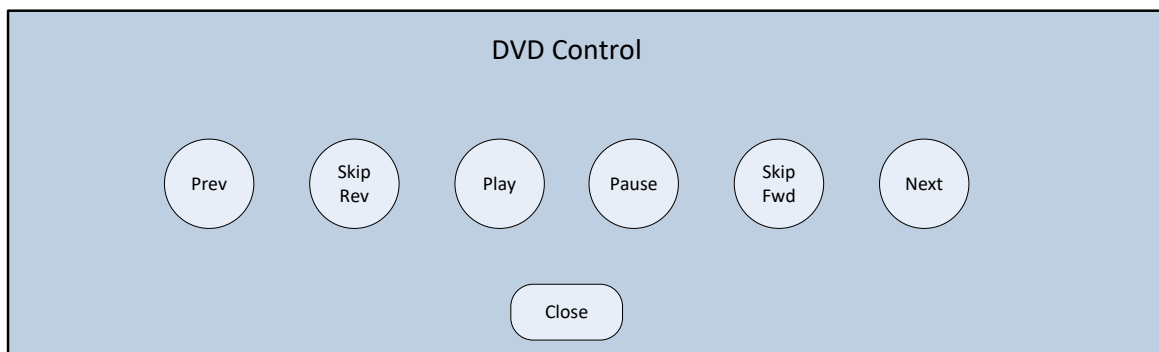


Figure 4. Sample GUI Layout - DVD Control Pop-up Page

3.0 CONTROL TOUCH PANEL GUI TEST PLANS

Control touch panel test plans shall be written in such a way as to force the operator to press every single button, operate every single slider, and perform every function within a room. The GUI test plan shall serve as a functional test for every sub-system, to include, all the hardware functions needed to control the room, as well as the





conductors and fibers within the system. The test plan shall be written in a sequential manner, so it leads the operator through the process of setting up the room in order to perform the test. Additionally, while leading the operator through the setup, they will become familiar with the operation of the test plan.

Each test plan line item shall use a line by line sign off process, with a line for the operator to place their initials. It is important that each item have its own line rather than one per page. Use of individual, initial lines will insure all individual functions are accepted and tested.

An Operator's Training Guide and Maintenance Test Routines (MTRs) shall be developed from the GUI test plan(s). Using the setup text from each test section, the operator can use the system to assist in failure detection and isolation. The Operator's Training Guide and Maintenance Test Routines shall use a narrative approach, instead of a matrix chart, as it forces the test operator to read and understand any notes the author needs, in order to properly conduct the test.

Appendix B provides a sample of an actual GUI test plan with a display control section, a Personal Computer (PC) workstation inputs section, and a Blu-Ray control section. These examples are not finalized products, they are provided to illustrate the use of a line by line, function by function sign off, with narratives. The narratives are used to explain the setup of the actual test function, the directive of the function is to perform the test and provide a positive affirmation that the test was passed.

4.0 OPERATOR TRAINING GUIDE

The Operator Training Guide shall be derived from the GUI Test plan text referenced in Appendix B. The Operator Training Guide is meant to train the operator in the use of the system. An index will be added to give the Operator Training Guide the ability to serve as a functional reference. Therefore, if an operator-in-training were to run the Operator Training Guide from start to finish, they will be able to understand the capabilities of the system as a whole and understand how all of the sub-systems function.

5.0 MTRS

MTRs will be defined to allow operators and support staff to troubleshoot and isolate problems within the system. Figure 5 shows a sample MTR.



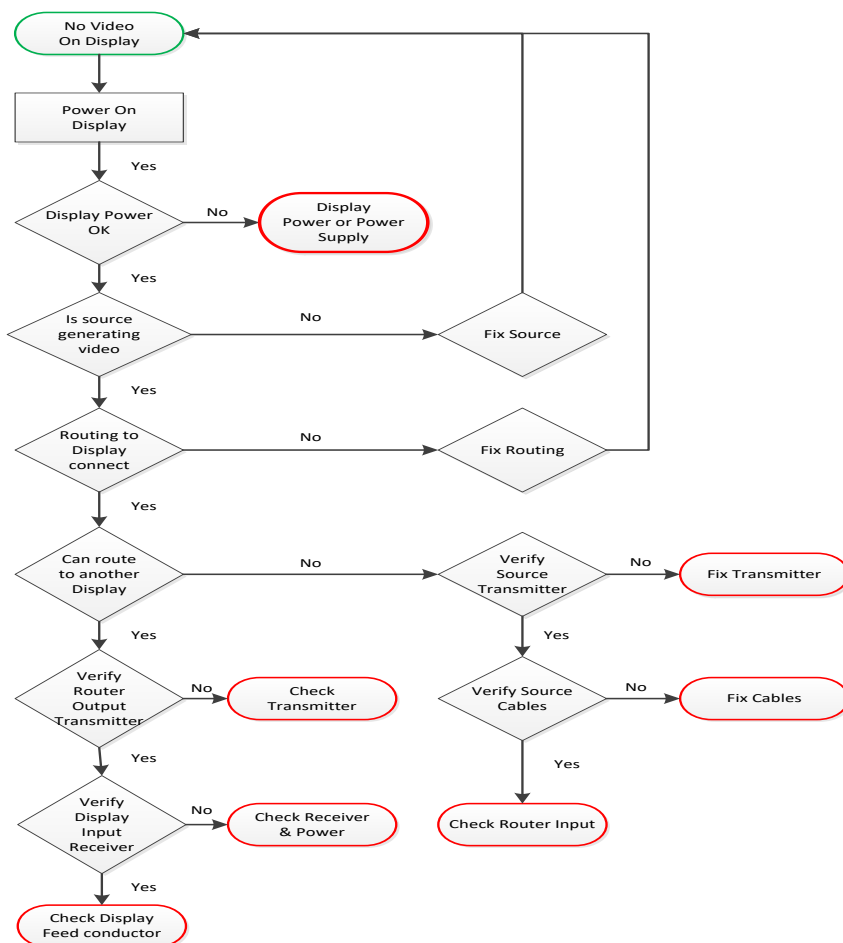


Figure 5. Sample MTR

6.0 OPERATOR/USER MACRO CAPABILITY

In an effort to cut down on third party software modifications throughout the service life of the VIS, this section describes a mechanism where the operator/user may construct “macro” command sequences, in order to create and execute custom non-hard coded room setup routines. There are two parts to this mechanism. The first part is a separate thread or routine running within the Crestron or AMX processor that listens to a specific Internet Protocol (IP) socket for commands. The second part relates to the Microsoft (MS)-Windows software application to allow the operator/user to create and execute the room setup macro command sequences.

Macro command sequences shall allow for powering displays on and off, routing input devices (workstations, DVD players, Television (TV) Tuners) to displays and speakers, change channels on tuners, etc. These macro command sequences shall NOT be able to route any source device to a transmit device or place calls on devices such as VTC Coder/Decoders (CODECs). Additionally, macro command sequences shall NOT be allowed to change room security levels.

Ideally, the MS-Windows software application will allow the operator/user to drag and drop power commands, sources, and destinations to a list which can be named and





saved as “macros” files for later recall. The operator/user will be able to load these macros and send them to the control system processor for execution. Figure 6 shows an example of a macro sequence. The operator/user will load the macro and send it to the A/V control processor for execution.

File Name: “Morning Update Brief.mcr”
<i>Display 1 power on</i>
<i>Display 2 power on</i>
<i>Input-Tuner 1</i>
<i>Output-Display 1</i>
<i>Output-Speakers</i>
<i>Input-Workstation 3</i>
<i>Output-Display 2</i>

Figure 6. Sample Operator Macro Guide

7.0 SOFTWARE DEVELOPMENT MILESTONES

VIS software development will run concurrent with EIP development. Figure 7 identifies the minimum content the software developer shall provide for the design review process.

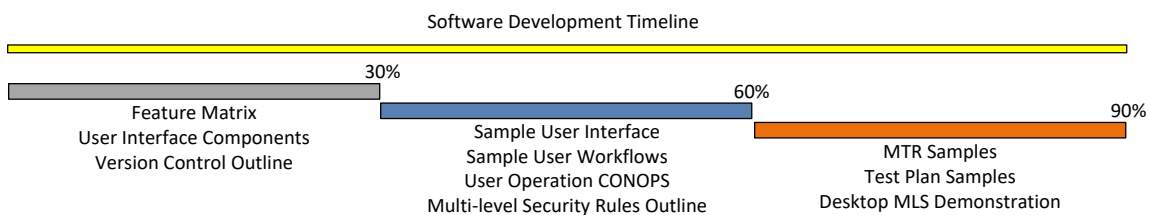


Figure 7. Sample Software Development Chart





APPENDIX A. REFERENCES

A-1.0 USG PUBLICATIONS

The following documents published by federal agencies were used to develop this report:

- a. Department of the Army Pamphlet (DA Pam) 25-40, Administrative Publications: Action Officers' Guide, 30 April 2002.
- b. Defense Information Systems Agency (DISA) Field Security Operations, Windows New Technology Security Checklist, Version 2, Release 1.8, 23 August 2002.
- c. Department of Defense Directive (DoDD) 5200.28, Security Requirements for Automated Information Systems, 21 March 1988.
- d. DoDD 5230.24, Distribution Statements on Technical Documents, 18 March 1987.
- e. Department of Defense Instruction (DoDI) 5200.40, Department of Defense Information Technology Security Certification and Accreditation Process Plan, 30 November 1999.
- f. Director of Information Systems for Command, Control, Communication, and Computers (DISC4) Message, Date-time Group (DTG) 1007337, SEP01, Subject: Information Assurance Certification Training Requirements. No date.
- g. Federal Information Processing Standards (FIPS) Publication 140-2, Security Requirements for Cryptographic Modules, 25 May 2001.
- h. Headquarters, Department of the Army (HQDA), Army Regulation (AR) 25-30, The Army Integrated Publishing and Printing Program, 21 June 1999.
- i. Military Standard (MIL-STD)-188-124B, Grounding, Bonding, and Shielding, 1 May 1998.
- j. National Computer Security Center (NCSC) Technical Guide-031, Version-1, Certification and Accreditation Process Handbook for Certifiers, July 1996.
- k. National Security Agency (NSA), NSA Guide to Securing Microsoft Windows New Technology Networks, Version 4.2, 18 September 2001.
- l. Policy Memorandum 00-1, AMSEL-IE-CO, Information Systems Engineering Policy, 17 March 2000.
- m. Public Law 100-235, Computer Security Act of 1987.

A-2.0 NON-USG PUBLICATIONS

The following documents published by civilian agencies were used to develop this report:

- a. Institute of Electrical and Electronics Engineers (IEEE), Standard 1100-2005 (Emerald Book), IEEE Recommended Practice for Powering and Grounding Electronic Equipment, 2006.
- b. IEEE Standard 802.11i-2004, IEEE Standard for Information Technology – Telecommunications and information exchange between systems – Local and metropolitan area networks – Specific requirements Part 11: Wireless Local Area





Network (LAN) (WLAN) Media Access Control (MAC) and Physical Layer (PHY) specifications – Amendment 6: MAC Security Enhancements.





APPENDIX B. SAMPLE TEST PLANS

B-1.0 SAMPLE TEST PLAN

B-1.1 Display Control Section

This section will test the power on and off of the eight (8) Liquid Crystal Display (LCD) display(s). See Table B-1.

- Press the “**Display Control**” button.
- Press the “**Power on**” button under the topic All Displays.
- Press and hold the “**Power off**” button under the Display 1 topic.
(Note: the button must be held for ~3-seconds to activate the power off function)
- Press and hold the “**Power off**” button under the Display 2 topic.
(Note: the button must be held for ~3-seconds to activate the power off function)

Table B-1. Display Control

	Y	N
The Display Control pop-up page is shown.		
All displays powered on.		
Display 1 powered off		
Display 2 powered off.		





B-1.2 PC Workstation Inputs

If all PC Workstations are not installed and functioning correctly this test section will fail.

If possible, ensure that each PC workstation is awake and displaying a signal, or the results of this test element will be inconclusive.

On all pop-up pages, that have more than one input selection available, there is an auto close feature that automatically dismisses the pop-up dialog after one device is selected for operator convenience.

This function will be turned on or off (default is off). In the lower left corner of the pop-up dialog is an auto close box. Pressing this box will enable the auto close function; a red check mark will indicate the function is enabled. Pressing the box again will remove the red check mark and turn off the auto close function.

- Press the **“Workstation”** button in the lower area of the UI.
- Press the check box labeled **“Auto Close”**.
- Press any of the **“Workstation”** buttons on the PC Workstation pop-up page.
- Press the **“Workstations”** button in the lower area of the UI.

Table B-2 shows the PC Workstation checklist.

Table B-2. PC Workstation

	Y	N
The PC Workstation pop-up page is shown.		
A red check mark appears in the check box.		
The PC Workstations pop-up page goes away.		
The PC Workstation pop-up page is shown.		





B-1.3 Blu-Ray Control Section

This section assumes the Blu-Ray disc, used in prior sections, is still in the player and playing the menu loop.

Due to the nature of Blu-Ray master authoring, many button functions, such as numeric buttons, Top Menu buttons, and Exit buttons will not be functional; therefore, these buttons are not tested in this test plan procedure.

- Press the “**Blu-Ray**” button on the UI.
- Press one of the display icons of an observable display.
- Press the “**Control**” button underneath the Blu-Ray button.
- Press the “**Down Arrow**” button.
- Press the “**Right Arrow**” button.
- Press the “**Up Arrow**” button.
- Press the “**Left Arrow**” button.

Select the “**Movie**” menu item to play the Blu-Ray disc content and allow time for the Federal Bureau of Investigation (FBI) Warning banner to expire and the movie to begin playing before continuing with this section.

- Press the “**Pause**” button.
- Press the “**Play**” button.
- Press the “**Stop**” button.
- Press the “**Play**” button.
- Press the “**FFWD**” button.
- Press the “**REW**” button.
- Press the “**Next**” button.
- Press the “**Previous**” button.
- Press the “**Menu**” button.
- Press the “**Close**” button.

Table B-3 shows the Blu-Ray Control checklist.



**Table B-3. Blu-Ray Control**

	Y	N
The hint text will indicate Blu-Ray.		
The Blu-Ray player signal is displayed on the display.		
The Blu-Ray player control page is displayed.		
The menu item drops down one menu selection. (Down Arrow)		
The menu item drops down one menu selection. (Right Arrow)		
The menu item rises up one menu selection. (Up Arrow)		
The menu item rises up one menu selection. (Left Arrow)		



**Table B-3. Blu-Ray Control (continued)**

The movie pauses or freezes at the location of the button press.		
The movie begins playing from the location of the button press.		
The movie stops playing at the location of the button press.		
The movie resumes playing from the location of the button press.		
The movie scans forward from the location of the button press.		
The menu scans in reverse from the location of the button press.		



**Table B-3. Blu-Ray Control (continued)**

The movie jumps to the next chapter and resumes playing.		
The movie jumps to the previous chapter and resumes playing.		
The disc main menu displays.		
The Blu-Ray player pop-up page is dismissed.		





APPENDIX C. VIS REQUIREMENTS MATRIX

C-1.0 VIS REQUIREMENTS MATRIX

The VIS Requirements Matrix will be provided separately.



VIS Requirements
Matrix.xlsx



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GLOSSARY. ACRONYMS AND ABBREVIATIONS

AR	Army Regulation
A/V	Audiovisual
B&D	Briefing and Display
C2F	Command and Control Facility
CODEC	Coder/Decoder
DA	Department of the Army
DISA	Defense Information Systems Agency
DISC4	Director of Information Systems for Command, Control, Communication, and Computers
DoD	Department of Defense
DoDD	Department of Defense Directive
DoDI	Department of Defense Instruction
DTG	Date-time Group
DVD	Digital Versatile Disc
EIP	Engineering Installation Plan
FBI	Federal Bureau of Investigation
FIPS	Federal Information Processing Standards
GUI	Graphical User Interface
HQDA	Headquarters, Department of the Army
I/O	Input/Output
IEEE	Institute of Electrical and Electronics Engineers
IP	Internet Protocol
IR	Infrared
LAN	Local Area Network
LCD	Liquid Crystal Display
MAC	Media Access Control
MIL-STD	Military Standard
MS	Microsoft
NCSC	National Computer Security Center
NSA	National Security Agency





Pam	Pamphlet
PC	Personal Computer
PHY	Physical Layer
SDP	System Design Plan
T&A	Tested and Accepted
TV	Television
UI	User Interface
USAISEC	U.S. Army Information Systems Engineering Command
USG	U.S. Government
VDS	A/V Distribution Subsystem
VI	Visual Information
VIS	Visual Information System
VTC	Virtual Teleconferencing
WLAN	Wireless Local Area Network

